Roll No. | | | | | | | | | | | Total No. of Pages : 03

Total No. of Questions : 13

B.Pharmacy (Sem.-8)

BIOSTATISTICS AND RESEARCH METHODOLOGY

Subject Code : BP801T

M.Code : 79764

Date of Examination : 28-04-2025

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION - A

1. **Answer briefly:**

   a) What are the characteristics of 'Frequency Distribution'?

   b) Give formulae for calculation of Median and Mode for a continuous series.

   c) Define the term 'Level of significance'.

   d) What are Type I and Type II statistical errors?

   e) What is the difference between Experimental and Observational studies?

   f) What is the need for 'Research'?

   g) What are the advantages of using R-Online Software in statistical analysis?

   h) What is Response Surface Plot?

   i) What are the measures of deviation from the Normal Distribution?

   j) Define Factorial Designs.

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SECTION - B

2. a) Explain various Non-probability Sampling Methods.

   b) Give a detailed account on Plagiarism.

3. **Write notes on :**

   a) Wilcoxon Rank Sum test

   b) Multiple Correlation Analysis

4. A study is conducted involving 10 patients to investigate the relationship and effects of patient's age and their blood pressure.

Age	B.P.
35	112
40	128
38	130
44	138
67	158
64	162
59	140
69	175
25	125
50	142

Find regression equation of data and predict the value of B.P. of Patient of age 60. Also test of Statistical significance at 5% level.

SECTION - C

5. What are the advantages of factorial Designs? Explain 2^3 factorial designs.

6. Write a note on central Composite Design.

7. Explain Blocking system for two level factorial.

8. What are the key features, advantages and limitations of SPSS in statistical analysis of clinical trial data?

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